

Solana Wallet Scanner

A smart, adjustable, evidence-backed research application. Built to run on your computer using free data allowances, without trading or a recurring software subscription.

PRODUCT BOUNDARY

Discover, reconstruct, screen, compare and monitor public Solana wallets. No trading, copy execution, paper trading, signing, wallet connection, custody or private keys.

The experience

Choose a preset. Scan a bounded batch. See which wallets match, miss or remain unresolved. Open any result to inspect the positions, costs and source evidence behind the numbers.

The engineering priority

Own the accounting and screening logic. Use providers for records and enrichment, not as an unquestioned source of profitable-wallet labels. Reduce scanning volume before reducing the standard of evidence.

Decision	V0.3 baseline
Deployment	Private local application; browser interface; no cloud server required.
Recurring subscription target	US\$0, within verified free allowances; development and computer costs excluded.
Data access	Native Solana RPC through Helius Free; GeckoTerminal enrichment; optional Birdeye Standard.
Local storage	SQLite and compressed evidence files; no database server needed.
Intelligence	Deterministic investigation rules and explanations; paid AI is not required.
Status	Researched design plus small offline reference controls. Not a built scanner or a wallet endorsement.

This revision supersedes the hosted/paid operating assumptions in v0.2. It preserves the strict research settings and scanner-only scope. The previous files remain historical references, not deployment instructions for this edition.

Prepared for Project Gmgn. Provider documentation checked on 2 October 2026. No API account, subscription, live scan, background monitoring or trading connection was created.

Decisions and changes

The free edition changes how much we process and where the application runs, not how profits or holding times are defined.

Area	Previous proposal	Updated decision
Application	Hosted private web service	Runs on one local computer; browser opens automatically.
Database	PostgreSQL server	SQLite by default; PostgreSQL remains a future multi-user option.
Evidence files	Cloud object storage	Local compressed records plus checksum manifests and backups.
Job processing	Hosted worker services	One durable local job queue; one database writer.
Discovery	Larger API-led batches	20 candidate addresses per batch; five deep audits as proposed starting limits.
AI	Optional cloud assistant	Template explanations by default; no paid model dependency.
Billing	US\$150-300/month allowance	US\$0 recurring subscription target; no upgrade or top-up path in the application.
Research quality	Versioned ledger and evidence	Unchanged. Partial records never become a historical match.

What the target does not promise

Free access is neither unlimited nor a guarantee that arbitrary historical wallets can be fully reconstructed. A current token-account list can miss accounts that were closed or whose ownership changed. Missing earlier costs or market observations can keep a wallet unresolved, even when all retrieved pages reconcile.

Decisions to freeze before coding

Keep the app single-user and Solana spot-research only. Preserve the existing strict preset without presenting it as an optimal profit formula. Use native RPC as the required data path. Make enhanced or premium endpoints unnecessary. Permit fewer than five matches, including zero.

SUCCESS CRITERION

One correct, reproducible wallet report under a free-account capability test comes before large-scale discovery. The number of attractive addresses is not an engineering acceptance criterion.

Simple to use, clear to inspect

The normal workflow is Scan > Results > Wallet Report > Compare > Watchlist. Technical controls belong behind an Advanced button.

Screen	Primary action	What must be visible
Home	Resume work	Last completed scans, stale reports, usage remaining and unfinished jobs.
Scan / Discover	Choose a preset or paste addresses	Scope, exact reporting window, free-data estimate and deep-audit cap.
Results	Filter and sort	Profit, median hold, position count, policy fit and data status separately.
Wallet Report	Show the calculation	Positions, transfers, fees, open holdings, methodology and source transactions.
Watchlist / Compare	Compare like for like	Same windows and metric definitions; material changes with evidence.
Settings	Save my preset	Thresholds, refresh frequency, workload caps, backup location and provider status.

First-run setup

The proposed launcher asks for a data folder, a manually obtained free Helius API key and an optional Birdeye Standard key. It runs a read-only capability and quota check, then offers one small calibration wallet. It never requests a wallet seed, a blockchain signature or a deposit.

Simple and advanced settings

Simple mode shows reporting period, minimum profit, minimum median hold and minimum completed positions. Advanced mode exposes the rest of the preset. Accounting methodology is a separate, versioned expert setting: changing dust or cost allocation triggers a rebuild, not a cosmetic filter update.

Truthful progress

Show stages, pages collected, records decoded, unresolved items and budget consumed. Do not show a fabricated percentage when the total history length is unknown. A paused scan keeps its checkpoint and explains what is needed next.

Proposed empty-state text: "No fully evidenced matches in this batch. Three wallets miss your settings; two need earlier cost records. View the reasons or inspect partial reports." Do not silently relax filters.

Local architecture

A modular application with an embedded database is sufficient for the bounded private pilot. No microservice cluster, cloud queue or mandatory container stack.



Component	Proposed implementation	Why
Interface	React + TypeScript, bundled assets	Clear settings and reports; no cloud UI dependency.
Research service	Python + FastAPI	Keep accounting, validation and reporting in one codebase.
Jobs	Persistent queue in SQLite	Resume after interruption; one job writer avoids needless infrastructure.
Ledger	Python int / Decimal; canonical strings	Exact quantities without binary floating-point accounting.
Storage	SQLite + local gzip JSON files	Portable single-user setup and retained source evidence.

SQLite is designed for embedded/local application uses. Its write model fits this proposal when writes are serialised. Monetary values must not be stored or aggregated as floating-point REAL values; use validated decimal strings and exact application arithmetic. [S16-S18]

Package a local launcher after the capability pilot. Pin dependencies and test on the selected operating system at implementation time. A cross-platform design is not a claim that installers have already been produced.

Free providers and access gates

Published free plans support a pilot. Every required method still needs an authenticated capability test before a live report can be called complete.

Source	Published position	V0.3 use
Helius Free	1M monthly credits; 10 RPC requests/second. Native archival methods currently listed at 1 credit each. [S01-S03]	Required: native address history, transaction details and account state. Start at 3 requests/second.
GeckoTerminal public	OpenAPI says approximately 10 calls/minute; support says 30. [S06-S07]	Free pool/token observations. Start at 8 calls/minute with backoff.
Birdeye Standard	US\$0; 30,000 compute units; 1 request/second. Selected historical prices are listed. [S08-S09]	Optional, disabled initially; enable only a tested endpoint with known CU cost.
Public Solana RPC	Rate-limited public infrastructure, not intended for production use. [S15]	Optional occasional cross-check, never the primary history dependency.

Do not depend on uncertain entitlements

getTransfersByAddress is documented as paid-plan-only. The indexed getTransactionsForAddress launch announcement restricted access to paid plans, while current pages have changed pricing and do not settle Free entitlement clearly enough for this design. Keep both out of the required free path; leave indexed history disabled until an explicit no-charge entitlement test is approved. [S02, S04-S05]

Account creation and billing

Use manual Dashboard Free signup, not a programmatic Agent signup that the provider says requires an anti-abuse payment. Free Mode does not buy credits or alter subscriptions. Provider-console settings and external usage still need checking; an app quota is not a contractual guarantee about the provider account. [S01]

The current English Helius credit reference supersedes the older 10-credit archival assumption for planning. Keep method costs and rates in a dated manifest, not hard-coded throughout the scanner. On a mismatch, pause and re-verify.

Collect records, prove the scope

Native RPC is the subscription-free foundation. Its account-specific nature is also the largest completeness risk we must design around.

Collection sequence

Freeze the network, UTC half-open window [start, end), finalised snapshot slot and methodology. Enumerate the wallet plus its current token accounts for both supported token programs. Page signatures with before/until; fetch unseen full transactions once and archive them before committing the next checkpoint. Native signatures and transaction metadata are documented separately. [S10-S13]

Inspect initialisation, closure and ownership-change instructions, pre/post token balances and supported inner instructions to discover additional historical accounts. Query those accounts too. Resolve actual mint identities and ownership at the event time, not from a current symbol or balance.

Do not overclaim discovery

A current token-account list is not a historical ownership index. Iterating until no additional accounts are discovered is useful but cannot prove that every previously closed, undiscovered account has been found. A completely hidden round trip can leave zero residual mismatch. Mark that scope limitation explicitly; require independent historical ownership evidence before a global completeness claim. [S04, S12]

Pagination and chronology

Keep successful and failed transactions. Detect repeated cursors, repeated page hashes, null transaction metadata, unsupported versions and missing time/order fields. Resolve same-slot ordering from supported block evidence where necessary; signature text is not chronological order. Filter dates locally to the exact half-open window. An error is never terminal evidence.

Opening cost recovery

Reach beyond 90 days when an included sale depends on older acquisitions. Derive earlier balances only from a verified anchor and complete intervening flows. A history or budget boundary is not a zero opening balance. A sale preceding the earliest observed purchase means unresolved earlier inventory, not automatically inventory before the report window.

COVERAGE CERTIFICATE

Record requested range, fetched scope, pages, terminal evidence, account set and discovery method, missing records, unsupported routes, unresolved basis and reconciliation results. Say "verified for stated scope" only when supported; never display an invented 100% completeness badge.

Accounting that can be checked

The scanner must explain how money moved before it claims the wallet has a profitable strategy.

Event	Analytical treatment
Normal swap	Reconstruct wallet-level economic exchanges; retain distinct swaps within one transaction.
Deposit or withdrawal	External capital movement, not trading profit. Value in-kind flows when required.
Token received without a purchase	Unknown cost basis until provenance is established; never assume a free winning trade.
Transfer between owned token accounts	Internal movement only when event-time ownership is evidenced.
Transfer to another wallet	Not a sale; do not automatically merge addresses based on heuristics.
Wrap/unwrap and refundable rent	Representation or asset-location change, not an automatic trading expense.
Failed transaction	Retain costs actually incurred; no invented successful swap.
LP, lending, leverage or unsupported route	Separate unsupported activity; block affected metrics rather than guess.

Exact, reproducible matching

Use raw integer quantities, actual mint decimals and analytical FIFO acquisition lots. FIFO is a reproducible research convention, not tax advice. Partial disposals consume only matched cost. Unsupported allocations remain visible as unknown; code must not silently discard them to improve returns.

Fee reconciliation

Attach attributable buy costs to basis, deduct exit costs once and show unallocated trading overhead separately. Include costs paid by the scoped wallet, not by a third-party sponsor. The network fee already includes prioritisation; do not subtract it twice. Embedded venue charges already reflected in received amounts also must not be deducted twice. [S13-S14]

For each asset: opening raw quantity + inflows - outflows = closing raw quantity. An exact balance check is necessary, but does not by itself prove ownership-history completeness or establish missing purchase costs.

Every ledger event links to a raw response, transaction signature and instruction/event path. Persist original observations separately from decoded interpretations so a parser correction can generate a traceable new report.

Profit, holding time and populations

Every metric gets one documented numerator, denominator, population and valuation method.

Period net realised trading P&L =
proceeds from in-window disposals
- matched acquisition basis (including allocated buy costs)
- exit costs - unallocated period trading overhead

Economic P&L = closing equity - opening equity
- external deposits + withdrawals

Report realised profit, open-inventory marks and total economic change separately. Realised ROI is return on disposed cost, not return on initial account equity. Use consistent event-time SOL and USD valuations; never divide all historical dollar gains by today's SOL price. Missing material prices keep the affected metric unresolved.

Metric	Defined population
Completed-position median hold	Episodes fully sold to zero whose closing sale falls in the selected window; first acquisition to final sale.
Position win rate	Net-positive completed episodes divided by all completed episodes, including breakeven outcomes.
Median position ROI	Median of net whole-episode ROI for that same completed cohort, not average sale ROI.
Rapid first-sale share	Share of completed episodes whose first positive economic sale occurs within five minutes.
Four profitable weeks	Four consecutive seven-day periods ending at report end; the other two days are outside this weekly test.
Largest token contribution	Highest aggregate positive realised P&L for one mint divided by positive total net period realised P&L.

Keep open and ambiguous positions visible

Show closed, open, transfer-interrupted and unresolved episode counts side by side. Unknown eligible episodes cannot disappear from a passing denominator. A tiny residual cannot become an undisclosed multi-day hold: strict zero closes remain primary; a separately named dust-adjusted diagnostic requires a new methodology version.

Also show time to first sale and time to 50%/90% of acquired units sold. A six-hour final exit can coexist with selling 90% after two minutes; these measure different behaviour. No holding metric establishes that a wallet is safe.

Free market marks are indicative, not an executable sale guarantee. Current liquidity cannot fill historical entry fields. No order quotes, trading replay or paper trading are part of this release.

Your settings, preserved and editable

The strict preset remains a research preference. Changing its thresholds cannot change accounting facts or erase missing data.

Control	Starting strict preset
Window / verification	30 days / 90 days; older cost history retrieved when needed
Net realised meme-trading profit	At least +5 SOL after identifiable costs
Realised ROI on disposed cost	At least +10%
Median completed-position ROI	At least +5%
Completed-position win rate	50%-85%
Median strict completed hold	1-72 hours
Completed positions	At least 50 in 30 days; 100 in 90 days
Traded meme-token mints	20-100; exclude settlement assets and spam
First sale within five minutes	No more than 10% of completed positions
Average buys / sells per episode	1-2 buys / 1-3 sells
Positive weeks	At least three of the last four seven-day periods
Largest token profit contribution	At most 25% of net realised period profit
Economic P&L	Positive with boundary values and external flows reconciled
Evidence gates	History, identity, basis, positions, fees, classification, valuation and findings review

What can change

Save alternative presets, adjust thresholds, choose display units, limit candidate batches and change refresh frequency. Preview the difference against cached data. A filter-only edit should normally make no provider calls; a period expansion or methodology change may require collection or rebuilding.

What cannot change

Missing evidence cannot be treated as a pass. Deposits are not profits. Fees and open holdings stay visible. The system cannot relabel averages as medians, quietly remove losing positions, enable trading or suppress an unsupported route.

A threshold miss is not suspicious behaviour. For example, trading 101 qualifying tokens rather than 100 is a policy miss, not a fraud finding. Do not automatically relax these values just to return five addresses.

Make it smart without paid AI

Intelligence comes from knowing what to investigate next, retaining evidence and explaining each decision clearly.

Smart behaviour	Proposed implementation
Spend effort where it matters	Resolve mint identity and history before expensive pricing. Prioritise missing evidence that can change a required decision.
Recover rather than guess	Backfill older acquisitions when basis is missing; keep the reason and incremental usage visible.
Stop wasteful work	Pause deeply ambiguous or over-budget wallets. Do not retry the same unsupported path indefinitely.
Reuse what we know	Cache raw transactions, token identities and priced intervals; re-filter metrics without downloads.
Explain every decision	Generate plain-language explanations from structured rule checks and evidence IDs.
Show sensitivity honestly	Preview threshold changes within the same dataset; do not optimise settings automatically to manufacture winners.
Monitor meaningful changes	Distinguish new behaviour from rolling-window effects and parser/policy changes.

No opaque legitimacy score

Keep three dimensions: policy outcome (MATCH/MISS/UNRESOLVED), evidence scope (verified/partial/unknown/stale), and specific findings. A known policy failure remains recorded even when another metric is unknown. A financial headline never overrides an evidence gate.

Examples of useful output

"Median hold is 43 minutes versus your one-hour minimum. This is a preference mismatch." "Two incoming allocations have unresolved cost; net profit remains unknown." "One token now contributes 34% of profit, up from 18%; inspect the contributing positions."

Optional language assistant, later

A local or user-selected model may propose schema-valid setting changes and explain existing reports. It cannot calculate the authoritative ledger, fetch arbitrary URLs or turn UNKNOWN into PASS. The whole product must work with AI disabled. External token metadata is untrusted content, a relevant prompt-injection risk. [S22]

MVP default: deterministic rules and template explanations. No cloud-model subscription, vector database or agent framework is required.

Discovery that fits a free budget

Free discovery should create a manageable research queue, not imply exhaustive coverage of every Solana trader.

Start from public addresses

Support pasted addresses and CSV/JSON imports containing public wallet identifiers only. Record the discovery source and date. A screenshot or third-party profit figure is a lead, not verified accounting. Prior conversational transaction examples are not golden test fixtures until independently retrieved.

Add bounded pool-based discovery

After accounting is reliable, sample selected Solana pools and validate candidate addresses through their raw transactions. GeckoTerminal documents pool endpoints and a recent-trades sample of up to 300 trades in the previous 24 hours. That is candidate discovery, not a 30-day wallet ledger. A sample may identify routers or authorities rather than the economic trader; resolve that before adding a wallet. [S06]

Avoid a leaderboard-only research process

Store each discovery cohort, selected pool universe and time. Use multiple sources and user-supplied candidates rather than only the largest token winners. Preserve misses and unresolved records so the team can measure the selection process, not just its surviving winners.

Workload control	Proposed starting value
Candidate addresses per batch	20
Deep-audit slots per batch	5
Active wallet audit	1, with bounded concurrent network reads
Soft transaction limit per wallet	10,000 unseen transactions; pause for review, not a silent truncation
Soft credit limit per wallet	20,000 estimated credits; continuation requires approval within the free cap
Discovery mode	On demand first; scheduled while app is running only

These values are engineering starting points, not capacity measurements. Prioritise existing watchlist completion ahead of speculative new discovery. Cache snapshots rather than repeatedly re-requesting the same pool or transaction.

No scraping behind authentication, bypassing limits, rotating accounts to evade quotas, or requiring a paid leaderboard. A narrow, accurately described discovery universe is acceptable for Free Local.

Quota-aware by construction

The app must know what a request could cost before sending it. Rate limits and monthly allowances are separate constraints.

Control	Proposed policy
Helius cycle ceiling	800,000 app credits against the published 1,000,000 allowance
Pause new discovery	At 600,000 app credits; prioritise reconciliation already in progress
Global stop	At the projected cap, counting outstanding reservations
Birdeye, if enabled	24,000 CU application cap; endpoint cost must be known
GeckoTerminal throttle	8 calls/minute initially; obey Retry-After and back off on 429
Auto spending	US\$0. No auto-upgrade, top-up, prepaid-credit purchase or payment flow.
Shared provider account	Reconcile external usage where available; retain headroom and dedicated-key guidance.

Reserve, dispatch, settle

Before each request, persist a reservation for its maximum metered cost. Atomically check all outstanding reservations. Release an undispached request; conservatively retain/charge a dispatched timeout until billing can be reconciled. Log retries and method-cost versions. An unknown method price, access failure or incompatible response stops that path.

Illustrative credit arithmetic

At the currently documented one credit per native getTransaction call, 20 wallets with 2,000 unseen transactions each require about 40,000 credits for transaction details alone. This excludes signatures, account discovery, backfill, valuations, failures and retries. It is arithmetic, not a throughput or completeness guarantee. [S02]

A ten-times cost stress case would be 400,000 credits for those detail calls. If a tariff or Free entitlement changes, reduce volume or pause. Do not switch to a paid endpoint automatically. Package-level caps follow the provider billing cycle, not an assumed calendar-month reset.

Progressive cost control

Measure credits per completed report, rate-limit frequency, unresolved history, cache reuse and local disk growth. Ask for explicit continuation when a wallet exceeds its soft limit. The same monthly hard cap still applies.

Free Mode protects the application path, not every action elsewhere in a provider account. Keep billing settings checked and no automatic top-ups. No repeated alternate-account creation to sidestep provider limits.

Local storage, jobs and resilience

Sleep, restarts and lost internet must pause work safely without inventing completion or losing evidence.

Storage layout

```
data/
  scanner.sqlite           # structured records and checkpoints
  evidence/<sha256>.json.gz # append-only by app convention
  exports/                 # explicit user-created reports
  backups/                 # consistent snapshots + manifests
  logs/                   # rotating, secrets-redacted logs
secrets: operating-system credential store, not this folder
```

SQLite WAL can permit readers while a writer operates, but this design keeps one writer and a local filesystem. Do not put the live database on a network share or sync its open files through a cloud-drive client. Use the SQLite backup interface for consistent snapshots. [S18-S19]

Durable job state

Each step has a scan ID, request identity, checkpoint, attempt count and lease. A response is archived before the checkpoint advances. Resume to the last durable step after a restart. Deduplicate transaction retrieval by signature while preserving distinct economic events within the same transaction.

Incremental refresh

Store a finalised high-water mark and retrieve overlapping activity on refresh. Do not assume an empty recently queried page proves old history is complete. Recalculate affected positions; retain prior report versions. Scheduled refreshes execute only while the local application runs.

Disk and backups

Proposed default: warn at a 10 GB evidence store and pause ingestion below 2 GB free disk. These are configurable operating guards, not storage forecasts. Back up the database with referenced evidence and preset versions; verify checksums and test restoration. Do not silently prune evidence supporting a published report.

Offline mode: open cached reports, change filters on cached verified metrics and compare saved snapshots. Refresh timestamps remain visible. A checksum-backed local archive is not tamper-proof against a user with filesystem access.

Contracts and module boundaries

All data crossing module boundaries carries explicit units, evidence status and versioned definitions.

Module	Input	Output
Provider gateway	Approved method + capability manifest + reservation	Archived response or typed access/quota/error state
Collector	Wallet + frozen window + checkpoint	Transactions, discovered accounts and scope certificate
Decoder	Raw transaction + account ownership	Mint-resolved economic flows or unsupported status
Accounting	Flows + supported valuations	Lots, positions, realised slices, inventory and reconciliation
Metric engine	Position/period populations	Typed metrics with numerator, denominator and evidence links
Policy engine	Metric snapshot + preset version	Per-rule PASS/FAIL/UNKNOWN; independent findings
Reports	Checks + evidence + report versions	Plain-language explanation, tables and safe exports

Core database entities

Wallets, scans, jobs, artifacts, transactions, account ownership intervals, economic events, acquisition lots, position episodes, metric results, findings, evidence links, market observations, versioned presets and watchlists. The pack supplies a draft SQLite schema for review; it is not a finished migration strategy.

Research-only API

Proposed routes create or inspect scans, pause/resume work, preview presets, retrieve reports and evidence, manage watchlists, inspect usage and export records. No arbitrary upstream proxy and no signing, order or transaction-submission route. Configuration mutations cannot bypass the server-side invariants.

Exact data representation

Pass raw quantities and financial decimals as strings across JSON and SQLite boundaries; validate with int/Decimal in Python. Use numeric computation rather than lexicographic TEXT sorting for financial filters. Booleans, NaN, Infinity and floating-point coercions are invalid metric inputs. [S17]

Keep runtime profile, provider-cost manifest, accounting methodology and screening preset versioned separately. Re-filter, rebuild and re-fetch are distinct actions with different costs.

Security without custody

No trading keys removes custody risk from the app, but provider credentials, private watchlists and local access still require protection.

Boundary	Required control
Blockchain permissions	Allowlist selected native read methods. Reject submission, bundles, airdrops, signing and simulation routes.
Network destinations	Static approved provider hosts and paths; validate redirects and TLS; no arbitrary URL proxy.
Local browser access	Bind to 127.0.0.1 only. Validate Host/Origin, restrict CORS, use a launch-session secret and CSRF protection.
Provider keys	Operating-system credential store; backend only; redact URLs, query strings and logs.
Untrusted metadata	Render as text; do not execute HTML or follow instructions from token descriptions.
Exports	Public addresses by default; explicit warning for labels/watchlists; prevent spreadsheet-formula injection in CSV text.
Updates and dependencies	Pin reviewed dependencies; verify packages; no remote script execution at runtime.
Backups	Exclude live credentials; preserve provenance; user-controlled local or encrypted backup destination.

Local is private, not invulnerable

No account login or wallet connection is required for the on-chain research. Nevertheless, another process or malicious webpage may attempt to access a local service. Session and origin controls must be tested. Remote access is explicitly outside V0.3; adding it requires a new authentication and network review.

Financial observations without safety labels

Report evidence of permissions, concentration and unusual flows. Solana permanent-delegate functionality is an example of a token capability worth detecting; it does not establish misconduct by itself. Lack of a detected finding is not proof of legitimacy, absence of insider activity or guaranteed sellability. [S21]

No GMGN API, Jupiter execution integration, funded test wallet, signing service, seed phrase storage, custody or paper-trading component is included. These omissions are architectural, not hidden switches waiting to be enabled.

Build plan with acceptance gates

Implement and independently inspect the hard data path before polishing the interface or scaling discovery.

Stage	Deliverable	Exit gate
0. Free capability proof	Free signup/access matrix, known transaction, account history and usage sample	Actual mint records and window coverage documented, or an explicit infeasibility/gap decision
1. One-wallet report	Collector, supported decoder, lots, positions and evidence output	Reproducible raw balances, costs and true medians where the data supports them
2. Ten-wallet pilot	Varied independent fixtures, unsupported-case handling and reconciliation	No silent omissions; reviewer signs expected accounting and scope limits
3. Local product	Launcher, settings, results, report, compare and export screens	Normal workflow requires no code editing or database administration
4. Monitoring and discovery	Budgeted batches, local refresh, resume, alerts and backups	Quota, sleep/restart, stale-data and restore tests pass
5. Optional assistant	Natural-language setting proposals and evidence summaries	Useful with AI disabled; no facts or permissions invented

Milestone ownership

One backend/data engineer can lead ingestion and accounting, with frontend work once the vertical slice is correct. An independent reviewer should verify the financial fixtures and edge cases. Security and installation acceptance are required before the owner relies on the application. These are proposed roles, not staffing commitments.

Stop/go decisions

Do not commission a large dashboard before Free access is proven. If native history cannot establish the required ownership or cost basis, the result is a scoped/partial report or a new data-source decision. Do not substitute paid access without consent and do not change the accounting standard to claim success.

The accompanying backlog is ordered by dependency, not by dates. No delivery-time or capacity estimate is asserted before the capability and accounting pilots produce measurements.

Tests that prevent attractive errors

Test correctness, completeness handling and free-only controls. A smooth interface cannot compensate for a wrong ledger.

Test group	Examples that must pass
Identity and collection	Same-symbol different mints; closed/changed token accounts; page repetition; unsupported versions; exact date cutoffs.
Accounting	Partial exits; re-entries; prior-window costs; deposits; transfer interruptions; rent/wrapping; failed fees; sponsored fees.
Metric populations	Median not mean; closed versus open cohort; within-window realisations; dust diagnostics; token-level concentration.
Free Mode	Unknown endpoint cost denied; paid methods denied; pending usage survives restart; 429 backoff; quota stop is not completion.
Local reliability	Sleep/resume; interrupted archive writes; disk-full behaviour; backups with evidence; restore then reproducible re-filter.
User controls	Valid thresholds save; impossible settings rejected; no invariant bypass; settings-only edits use cached metrics.
Security	Submission attempts blocked; hostile origins rejected; keys redacted; malicious token HTML inert; safe CSV export.

Independent fixtures

Obtain raw transactions from primary records and independently calculate a small expected ledger. Include losses and unresolved cases, not only winners. Earlier AI-generated wallet summaries are not valid expected-output fixtures. Keep live account capability tests separate from synthetic unit tests.

Observable operating measures

Record known-versus-unresolved positions, retrieval gaps, decoder coverage, fee reconciliation, API units per report, cache-hit rate, provider failures, disk usage and report age. Do not publish a precision score that has not been calibrated against independently reviewed data.

The pack contains offline reference policy and quota controls with executable synthetic tests. They do not implement transaction decoding, provider authentication or a working app. Their exact executed test result is recorded in VALIDATION.md, not extrapolated to live scanner reliability.

Operating plan and practical limits

The target is no recurring software/data subscription, using your existing computer and workloads that fit confirmed free access.

Item	Recurring subscription target	Qualification
Primary RPC	US\$0	Helius Dashboard Free; actual access and usage must be checked. [S01]
Market enrichment	US\$0	GeckoTerminal and optional Birdeye Standard within their free restrictions. [S06, S08]
Database and local hosting	US\$0	Embedded SQLite; no cloud host or database service. SQLite is public domain. [S20]
AI explanations	US\$0 required	Deterministic templates; a model is not necessary.
Development and maintenance	Not included	Implementation, reviews, packaging and future decoder work still have a cost.
Hardware, internet and backups	Not included	Use existing resources; additional storage or hardware is optional and not assumed free.

Normal use

Open the app and check provider/cost-manifest status. Load a preset and a small candidate batch. Review the predicted budget before starting. Read complete and partial results distinctly. Add useful addresses to the local watchlist. Back up successful reports and their evidence. Cached results remain readable without network access.

When something breaks

Access denied: disable that method, preserve the job and show the entitlement gap. Rate limited: respect backoff. Budget exhausted: pause until the cycle resets or the user reduces workload. Missing price or older acquisition: keep affected results unresolved. Unsupported transaction: preserve raw evidence and request a reviewed decoder addition.

What is deliberately not promised

No whole-chain continuous scan, guaranteed five matches, complete historical ownership discovery for every address, fully executable portfolio valuation, always-on alerts while the computer is off, or a future provider free-plan guarantee. Supporting one genuinely complete report is better than displaying hundreds of unverifiable rankings.

The system should never convert "we reached the free limit" into "this wallet passed." Reduce batch size and refresh frequency first; preserve the evidence standard.

Handoff and first implementation task

This package replaces the paid/hosted baseline with a concrete Free Local engineering plan. It is ready for capability-pilot implementation, not production use.

Included deliverables

The PDF, editable Word and Markdown report; architecture source; free-runtime profile; strict preset and schemas; draft local data/API contracts; accounting rules; provider capability checklist; prioritised backlog; acceptance cases; source register; and small offline policy/quota reference controls.

The first task for the engineer

Using a manually created Free account, retrieve an independently verifiable Solana transaction and a low-complexity wallet history. Demonstrate real token mint IDs, supported transaction versions, address/token-account paging, exact time filtering and usage accounting. Do not use a private key and do not invoke any paid or transactional method.

Required pilot output

Produce a dated capability manifest, raw-response evidence archive, budget measurements, owned-account scope explanation and a draft ledger with every unresolved acquisition or fee called out. Decide whether the free-native path supports a complete report for that particular case. Repeat with a deliberately difficult wallet before designing broad discovery.

Validation boundary

Public documentation was researched; free-provider accounts were not authenticated here. No live wallet history, trading result, detector accuracy, desktop installation, UI or production security was validated. The included reference tests run offline on synthetic inputs only. No subscriptions, background jobs or blockchain actions were created.

ENGINEERING DEFINITION OF DONE

An owner can open the app, choose settings, run a bounded free scan, understand why each result matches/misses/remains unknown, inspect the source evidence and safely resume later. No hidden payment dependency and no trading capability.

The original goal of finding useful slower-holding traders remains a research objective. The application must be equally willing to show a loss, a policy mismatch or an unresolved wallet when that is what the evidence supports.

Primary source register

Primary documentation supports the provider and protocol facts. Architecture choices, quotas and screening thresholds are proposed engineering decisions, not provider guarantees.

S01 Helius plans

<https://www.helius.dev/docs/billing/plans>

Free: 1M monthly credits and 10 RPC requests/s. Dashboard signup differs from paid anti-abuse Agent signup. No account access tested.

S02 Helius credit schedule

<https://www.helius.dev/docs/billing/credits>

Current English reference lists native historical calls at 1 credit; enhanced history is separately metered; getTransfersByAddress requires a paid plan.

S03 Helius rate limits

<https://www.helius.dev/docs/billing/rate-limits>

Endpoint-specific rate limits; treat published maxima as ceilings, not app targets.

S04 Helius indexed history

<https://www.helius.dev/docs/rpc/gettransactionsforaddress>

Documents indexed token-account coverage and time filters; current Free entitlement not authenticated.

S05 Helius original indexed-history launch

<https://www.helius.dev/blog/introducing-gettransactionsforaddress>

Older announcement restricts indexed history to paid plans. Do not infer current Free access from missing restrictions elsewhere.

S06 GeckoTerminal OpenAPI specification

<https://api.geckoterminal.com/docs/v2/swagger.json>

Approx. 10 public calls/minute. Pool-trade sample: latest 300 in 24 hours. Pool and token endpoints are enrichment and candidate seeds, not complete wallet history.

S07 CoinGecko support: GeckoTerminal API

<https://support.coingecko.com/hc/en-us/articles/22612838274841-Does-GeckoTerminal-have-an-API>

Support still states 30 calls/minute. Conflict with current OpenAPI; app starts at 8/minute and backs off.

S08 Birdeye pricing

<https://docs.birdeye.so/docs/pricing>

Standard: free, 30,000 CU and 1 request/s; limited access; optional, disabled by default.

S09 Birdeye package access

<https://docs.birdeye.so/docs/data-accessibility-by-packages>

Selected historical-price methods listed for Standard; token/time coverage and actual account access require testing.

S10 Solana address signatures

<https://solana.com/docs/rpc/http/getsignaturesforaddress>

Native address pagination with before/until; account-specific records, not an owner-wide index.

S11 Solana transaction details

<https://solana.com/docs/rpc/http/gettransaction>

Full transaction metadata retrieval and version support; null results are not zero-value trades.

Primary source register / continued

Primary documentation supports the provider and protocol facts. Architecture choices, quotas and screening thresholds are proposed engineering decisions, not provider guarantees.

S12 Solana owned token accounts

<https://solana.com/docs/rpc/http/gettokenaccountsbyowner>

Current owned token-account enumeration, not a historical ownership snapshot.

S13 Solana transaction JSON structures

<https://solana.com/docs/rpc/json-structures>

Mint identities, raw quantities, fees, account balances and inner instructions where available.

S14 Solana fee structure

<https://solana.com/docs/core/fees>

Network fee comprises base and optional prioritisation; do not double-subtract priority fees.

S15 Solana public endpoints

<https://solana.com/docs/references/clusters>

Public endpoints have rate limits and are not intended as production infrastructure.

S16 SQLite appropriate uses

<https://sqlite.org/whentouse.html>

Embedded local application use; single-writer design is appropriate for this bounded single-user proposal.

S17 SQLite floating-point caveats

<https://sqlite.org/floatingpoint.html>

Binary floating point is approximate; ledger uses canonical strings and Python Decimal/int.

S18 SQLite write-ahead logging

<https://sqlite.org/wal.html>

WAL concurrency and local-filesystem constraints; serialize writes and do not live-sync the database.

S19 SQLite backup API

<https://sqlite.org/backup.html>

Use supported snapshots rather than copying an open database file alone.

S20 SQLite licence

<https://sqlite.org/copyright.html>

SQLite is public domain; no database service subscription is required.

S21 Solana permanent delegate

<https://solana.com/docs/tokens/extensions/permanent-delegate>

Example token capability to observe; no automatic claim of fraud or proof of sellability.

S22 OWASP prompt injection

<https://genai.owasp.org/llmrisk/llm01-prompt-injection/>

Untrusted metadata can carry indirect instructions. Optional AI has no network or authority bypass.